Auteur: Hendrik De Bruyn
Studierichting: Informatica
Oefeningenreeks 2D nr. 12.11

Opgave: Bereken de limiet $\lim_{x\to\infty}(x+3-\sqrt{x^2+6x})$

Oplossing:

$$\lim_{x\to-\infty}(x+3-\sqrt{x^2+6x}) = -\infty-(+\infty) = -\infty$$

$$\begin{aligned}
\lim_{x\to+\infty}(x+3-\sqrt{x^2+6x}) &= \lim_{x\to+\infty}\frac{(x+3-\sqrt{x^2+6x})(x+3+\sqrt{x^2+6x})}{(x+3+\sqrt{x^2+6x})} \\
&= \lim_{x\to+\infty}\frac{x+3)^2-x^2+6x}{(x+3+\sqrt{x^2+6x})} \\
&= \lim_{x\to+\infty}\frac{9}{(x+3+\sqrt{x^2+6x})} \\
&= \frac{9}{+\infty} \\
&= 0
\end{aligned}$$

References